K0WLY Two-Way CW Keyer

LilyGO T-Display S3 AMOLED

Operator's Manual — Preliminary

Version 1.6.0

K0WLY — Saratoga Springs, Utah — Grid DN40

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1. Overview

The K0WLY Two-Way CW Keyer is an ESP32-S3 based iambic keyer on the LilyGO T-Display S3 AMOLED (v1.6.0).

- Iambic Mode A, Iambic Mode B or Ultimatic keyer, 5–40 WPM, straight key mode
- Farnsworth spacing — character speed plus overall (effective) speed, e.g. 20/10
- Key output via PC817 optocoupler (GPIO12)
- Iambic DIT/DAH output (GPIO40/41) for radios with built-in keyers
- Paddle reverse and keyer mode (A/B/Ultimatic) via GPIO16
- Adjustable sidetone 400–900 Hz, logarithmic volume
- Head copy delay 0–3s or Audio Only mode
- Word gap spacing (off or 4–9 dits)
- WiFi AP (K0WLY-XXXX) with web interface for settings and file management

- Text file playback with pause/resume, drives iambic output
- Automatic ESP-NOW peer-to-peer pairing, two-way CW
- Full Morse decoding: A–Z, 0–9, punctuation, prosigns

2. Display Layout

Zone	Location	Content
Header Bar	Top 32px	WPM, Hz, DLY, VOL, GAP, A/B/U/SK, SOLO/DUAL
TX Line	Upper middle	Outgoing chars (green)
RX Line	Lower middle	Incoming chars (cyan)
Status Area	Bottom 96px	Pot mode, value, peer MAC, unit ID, version

2.1 Header Bar

- WPM or WPM/FARN — character speed / overall (effective) speed
- Hz — sidetone frequency
- DLY — head copy delay or AO
- VOL — volume percentage
- GAP — word gap setting
- A / B / U / SK — Iambic Mode A, Iambic Mode B, Ultimatic, or Straight Key
- SOLO / DUAL — ESP-NOW peer status

3. Controls and Connections

3.1 GPIO Pin Assignments

Function	GPIO	Notes
DIT Paddle Input	GPIO11	Active LOW, ext 10kΩ pull-up
DAH Paddle Input	GPIO10	Active LOW, ext 10kΩ pull-up
Key Out	GPIO12	HIGH=keyed, PC817 optocoupler
Sidetone	GPIO13	PWM → 470Ω → 2N4401 → speaker/headphones
Pot (ADC)	GPIO14	Wiper of 10–20kΩ pot
Key Mode	GPIO15	SPST: open=iambic,

		closed=straight key
Paddle Reverse / Keyer Mode	GPIO16	Short=dit/dah swap, Long=cycle Mode A/B/U
Pot Mode Select	GPIO39	PBNO: short=cycle, long=edit
Iambic DIT Out	GPIO40	Active HIGH, PC817 optocoupler to radio DIT
Iambic DAH Out	GPIO41	Active HIGH, PC817 optocoupler to radio DAH

3.2 GPIO16 Button Functions

Press	Action
Short press	Toggles dit/dah swap (also available on web page)
Long press (1 sec)	Cycles keyer mode A → B → U (Ultimatic) → A — header shows A, B or U

3.3 Keyer Modes

Mode A (default) — when both paddles are released during an element, the keyer finishes that element and stops.

Mode B — when both paddles are released during an element, the keyer sends one additional opposite element before stopping. Preferred by some operators for smoother squeeze keying.

Ultimatic (U) — when both paddles are squeezed, the paddle pressed most recently wins and repeats instead of alternating. Release it and the keyer goes to the paddle still held, or stops if none is. A new tap of the opposite paddle during an element is remembered and sent next. A paddle that is simply held down is not remembered, so letting go of both paddles stops the keyer after the current element.

> **NOTE:** *Keyer mode (A/B/U) is saved to flash and restored on power cycle.*

3.4 Key Output (GPIO12)

Drives a PC817 optocoupler to key the transceiver. Active HIGH. Never connect GPIO12 directly to a radio.

3.5 Iambic DIT/DAH Output (GPIO40/41)

For radios with built-in keyers. Active HIGH (GPIO HIGH = LED on = radio keyed). Both outputs driven during file playback. Paddle reverse applies to outputs.

- Tip = DIT (GPIO40 optocoupler)
- Ring = DAH (GPIO41 optocoupler)
- Sleeve = GND

NOTE: *Set the radio's WPM to match the keyer's CHARACTER speed. Farnsworth gaps are transparent to the radio.*

3.6 Audio Circuit

Speaker — direct, NO coupling capacitor:

3.3V → Speaker → Collector / 2N4401 / GPIO13 → 470Ω → Base / Emitter → GND

Headphone — with coupling cap:

3.3V → 1kΩ → Collector / 2N4401 #2 / Emitter → GND. Collector → 10kΩ volume pot → 47µF bi-polar → 100Ω → 3.5mm tip.

4. Parameter Adjustment

4.1 Pot Mode Button

Press	Action
Short press	Cycle: WPM → FREQ → DELAY → VOL → GAP → WPM
Long press (1s)	Enter/exit edit mode
Short press (in edit)	Exit edit and advance to the next parameter (in WPM, the first short press moves to the Farnsworth step). Values save as you adjust

4.2 Parameter Ranges

Parameter	Range	Notes
Char Speed	5–40 WPM	PARIS standard
Farnsworth	4 WPM – char speed	Overall (effective) speed
Frequency	400–900 Hz	Each unit independent
Head Copy Delay	0.0–3.0s or AO	Top 10% = Audio Only
Volume	2–100%	Logarithmic
Word Gap	OFF or 4–9 dits	Bottom 20% = OFF

5. WiFi Operation

5.1 Access Point

Each unit creates K0WLY-XXXX (XXXX = last 4 hex of MAC). No password. Browse to http://192.168.4.1.

5.2 Web Interface

- Settings tiles — Char Speed, Farnsworth, Frequency, Volume, Head Copy Delay and Word Gap each have − and + buttons
- Keyer Mode tile — tap Iambic A, Iambic B or Ultimatic
- Paddles tile — tap Dit-Dah or Dah-Dit (same as the hardware paddle reverse button)
- Changes are saved on the keyer, and the page picks up changes made with the knob and buttons every 2 seconds
- Farnsworth shows OFF when it equals the character speed
- On the web page, character speed steps 38 ↔ 40 (39 is skipped, since 39 and 40 WPM share the same 30 ms dit). The knob and buttons can still select 39
- Practice Files — Play (▶️) and Delete (✕) per file
- Stop Playback — immediate
- Pause/Resume — resumes from same position
- Upload Text File — .txt files only

5.3 Text File Playback

- TX line clears on start, characters appear as they sound
- A newline counts as one word gap; blank lines and a newline right after a space add nothing extra. Spaces you type are always kept
- KEY OUT and iambic outputs driven during playback
- Elements transmitted to peer over ESP-NOW

5.4 ESP-NOW Peer

Units auto-discover on channel 1. Header shows DUAL (green) when paired.

6. Firmware Information

Item	Details
Platform	ESP32-S3 (LilyGO T-Display S3 AMOLED)

Framework	Arduino via PlatformIO
Board Definition	lilygo-t-amoled (custom boards/ folder)
WiFi	AP+STA: ESPAsyncWebServer + ESP-NOW
File Storage	LittleFS on internal flash
Keyer Task	FreeRTOS Core 1, 1ms resolution
Keyer Modes	Mode A, Mode B and Ultimatic (GPIO16 long press)
NVS Version	4
CW Standard	PARIS (ITU-R M.1677-1)
License	CERN-OHL-W v2 / CC BY 4.0

7. Default Settings

Setting	Default
Speed	20 WPM
Farnsworth	Off
Sidetone Frequency	700 Hz
Head Copy Delay	0.0 seconds
Volume	64%
Word Gap	OFF
Keyer Mode	Mode A
Paddle Reverse	Normal
AP SSID	K0WLY-XXXX (unique per unit)

8. Troubleshooting

Symptom	Likely Cause	Fix
Display blank on boot	Crash	Hold BOOT, press RST, reflash
Always SOLO	Peer out of range	Both units on channel 1
Wrong characters	Paddles reversed	Short press GPIO16 or web swap
No sidetone	Volume at min	Enter VOL edit, increase
No speaker audio	Coupling cap installed	Remove cap
Radio not keying	Optocoupler issue	Check GPIO12 circuit

G90 wrong speed	G90 WPM mismatch	Set G90 WPM = char speed
K0WLY-XXXX not visible	AP not started	Check serial monitor
Files not showing	LittleFS issue	Reflash with correct board

Morse Code Reference

Letters:

A ·−	B −···	C −·−·	D −··	E ·	F ··−·
G −−·	H ····	I ··	J ·−−−	K −·−	L ·−··
M −−	N −·	O −−−	P ·−−·	Q −−·−	R ·−·
S ···	T −	U ··−	V ···−	W ·−−	X −··−
Y −·−−	Z −−··				

Numbers:

0 -----	1 ·----	2 ··---	3 ···--	4 ····-	5 ·····
6 −····	7 −−···	8 −−−··	9 −−−−·		

Punctuation:

. period ·−·−·−	, comma −−··−−	? question ··−−··	/ slash −··−·	(open −·−−·	) close −·−−·−
: colon −−−···	; semi −·−·−·	- hyphen −····−	' apos ·−−−−·	" quote ·−··−·	@ at ·−−·−·
& amp ·−···	! excl −·−·−−	_ under ··−−·−	$ dollar ···−··−		

Prosigns:

+ AR ·−·−· end msg	= BT −···− break	~ SK ···−·− end QSO	& AS ·−··· wait		

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